

# Week 6 – Workshop

## Magnetics: Anomalies and Susceptibility



Geophysics 3A: Potential Fields & Geothermics

July 6, 2026

**Duration:** 3 hours

**Setting:** Workshop room (disc magnets, iron filings, susceptibility meter)

### Preparation

**Pre-reading:**

**Lecture videos:**

**Notes:**

### Purpose

This workshop builds physical intuition for magnetic fields and their geological sources. Using dipole demonstrations, rock susceptibility measurements, and a case study from Antarctica, students contrast magnetics with gravity (vector nature, polarity, temperature dependence) and connect magnetic structure to crustal composition and thermal state.

### Learning Outcomes

By the end of this workshop, students should be able to:

- Visualise magnetic fields generated by dipoles and combinations of sources
- Relate magnetic susceptibility to rock type and geological process
- Predict magnetic anomaly shapes from simple geological geometries
- Contrast magnetic anomalies with gravity anomalies for the same structures
- Explain the concept of Curie depth and its relationship to thermal structure
- Link source depth to anomaly wavelength and spectral content

### Session Flow (180 min)

#### 0–10 min: Introduction and framing

- How magnetics differs from gravity (vector nature, polarity, temperature dependence)
- Overview of the three activities and how they connect

#### 10–35 min: Activity A: Magnetic fields from dipoles

- Iron filings and disc magnets
- Vertical vs horizontal dipole orientation
- Stacked magnets and multiple sources

#### Learning emphasis:

- Dipole field geometry

- Orientation-dependent symmetry and polarity
- Intuition for induced magnetisation

**35–55 min: Activity B: Whole-class demo: Magnetic susceptibility of rocks**

- Susceptibility measurements of igneous, metamorphic, and sedimentary rocks
- Effects of alteration, metamorphism, and weathering

**Learning emphasis:**

- Susceptibility as a material property
- Why similar compositions can have very different magnetic behaviour

**55–95 min: Activity C: Magnetic anomalies vs gravity**

- Reasoning-based sketching of anomalies for idealised geometries
- Direct comparison with gravity anomalies

**95–105 min: Break****105–160 min: Activity D: Curie depth and magnetic crust**

- Curie temperature and magnetic crustal thickness
- Depth–wavelength relationships
- Antarctica case study
- Back-of-the-envelope heat-flow estimates

**160–180 min: Synthesis and wrap-up**

- What magnetism can and cannot tell us
- Connection to upcoming Fourier/spectral methods

## Activity A

### Visualizing Magnetic Fields

#### Purpose

This activity develops physical intuition for magnetic fields as vector fields generated by dipoles. You will explore how field geometry depends on source orientation, strength, and proximity.

#### Materials

- Disc magnets
- Paper sheets
- Iron filings

#### Part A: Single disc magnet — vertical orientation

Place a disc magnet flat on the table and sprinkle iron filings over a sheet of paper on top.

1. Describe the symmetry of the field pattern.
2. Where are the field gradients strongest?
3. If this magnet were buried at depth, where would you expect the strongest anomaly?

**Sketch:**

#### Part B: Single disc magnet — horizontal orientation

Stand the same magnet on its edge and repeat the experiment.

1. How does the field symmetry differ from Part A?
2. Where are positive and negative lobes located?
3. How would this differ from a gravity anomaly produced by the same body?

**Sketch:****Part C: Stacked disc magnets**

Stack several disc magnets and repeat Parts A and B.

1. How does stacking affect field strength?
2. Does stacking change anomaly shape or only amplitude?
3. What geological parameter does stacking mimic?

**Reflection**

Why are magnetic field patterns sensitive to orientation, while gravity fields are not?

| Sample | Rock type | Geological context | Susceptibility (SI) | Notes |
|--------|-----------|--------------------|---------------------|-------|
| 1      |           |                    |                     |       |
| 2      |           |                    |                     |       |
| 3      |           |                    |                     |       |
| 4      |           |                    |                     |       |
| 5      |           |                    |                     |       |
| 6      |           |                    |                     |       |
| 7      |           |                    |                     |       |
| 8      |           |                    |                     |       |

## Activity B

### Magnetic Susceptibility of Rocks

#### Purpose

This activity links magnetic susceptibility to mineralogy and geological processes such as alteration, metamorphism, and weathering.

#### Key Reminder

Magnetic susceptibility ( $\chi$ ) describes how strongly a material becomes magnetised:

$$\mathbf{M} = \chi \mathbf{H}$$

#### Samples and Observations

##### Paired Comparisons

##### Peridotite vs Serpentinite

How does serpentinisation lead to high magnetic susceptibility?

##### Fresh Gabbro vs Altered Basalt

Why can alteration dramatically reduce susceptibility in mafic rocks?

**Granite or Sedimentary Protolith vs Metamorphosed Equivalent**

Which minerals control susceptibility in felsic or sedimentary rocks?

**Synthesis**

1. Why is susceptibility not a simple proxy for bulk composition?
2. Which geological processes increase susceptibility?
3. Which decrease it?

**Reflection**

Explain why petrology is essential for interpreting magnetic anomalies.

## Activity C

### Magnetic Structures

#### Purpose

This activity mirrors previous gravity workshops. You will reason about magnetic anomaly shapes and compare them directly with gravity anomalies from the same structures.

Assume:

- Uniform susceptibility
- Induced magnetisation only
- Constant inducing field

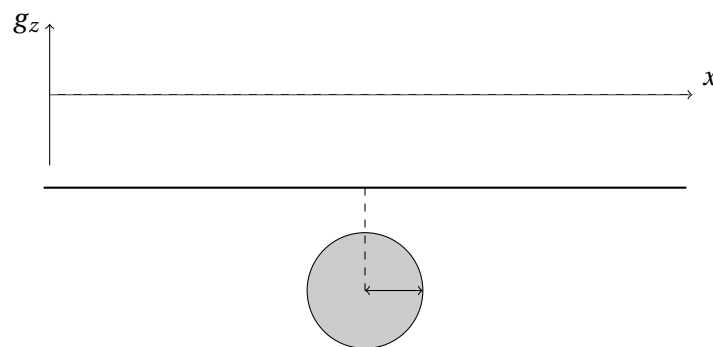
#### Instructions

For each structure:

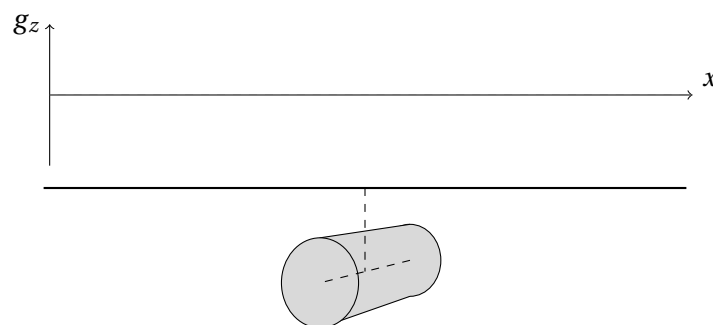
1. Sketch the expected vertical magnetic anomaly
2. Indicate sign, symmetry, width, and strongest gradients
3. State how it differs from the gravity anomaly

#### Structures

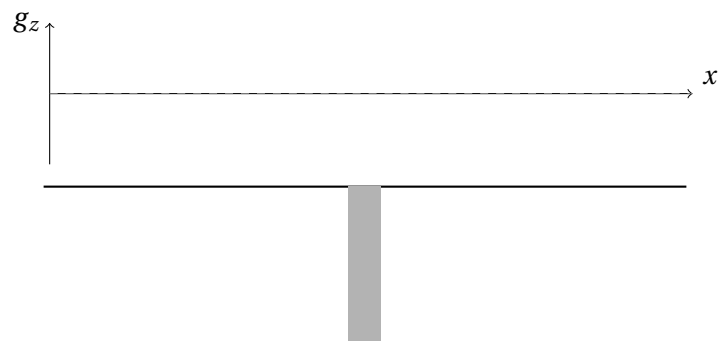
##### A) Buried Sphere (3D)



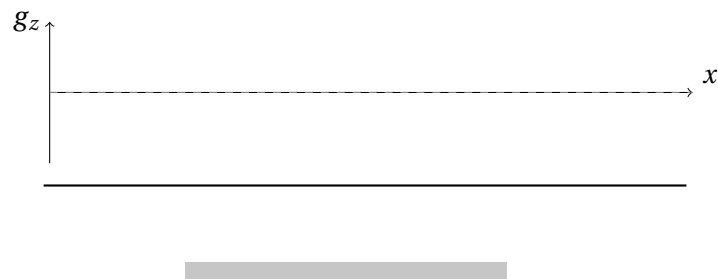
##### B) Buried Cylinder (2D; infinite strike)



##### C) Vertical Slab



#### D) Finite-Extent Horizontal Slab



### Vertical vs Horizontal Inducing Field

Repeat one case assuming a horizontal inducing field.

1. How does the anomaly shape change?
2. Which case appears more gravity-like?
3. Why does gravity not behave this way?

### Reflection

Give two reasons why magnetic interpretation is more non-unique than gravity.



## Activity D

### Curie Depth and Magnetic Crustal Thickness

#### Purpose

This activity explores how temperature limits crustal magnetisation and why deep magnetic structure is long-wavelength.

#### Part A: Curie Depth Concept

1. What mineral dominates crustal magnetisation?
2. What happens above the Curie temperature?
3. Is Curie depth a thermal or compositional boundary?

#### Part B: Depth and Wavelength

Given several magnetic profiles:

1. Which correspond to shallow sources?
2. Which to deep Curie depth?
3. How does depth affect short wavelengths?

#### Part C: Antarctica Case Study

Consider the magnetic and crustal structure of Antarctica (Fig. 1).

Compare:

- Magnetic anomalies
- Seismic crustal thickness
- Magnetic crustal thickness

Why do seismic and magnetic thickness estimates differ?

The first-order geology of Antarctica is characterized by a stable (cratonic) region in East Antarctica and a Cenozoic rift in West Antarctica (Fig. 2).

#### Part D: Heat-Flow Estimate

Assume:

$$T_C = 570^\circ\text{C}, \quad k = 2.8 \text{ W m}^{-1}\text{K}^{-1}$$

$$q = k \frac{T_C}{z}$$

Estimate heat flow for a rift and a craton.

#### Reflection

Explain why Curie depth makes magnetism sensitive to thermal structure.

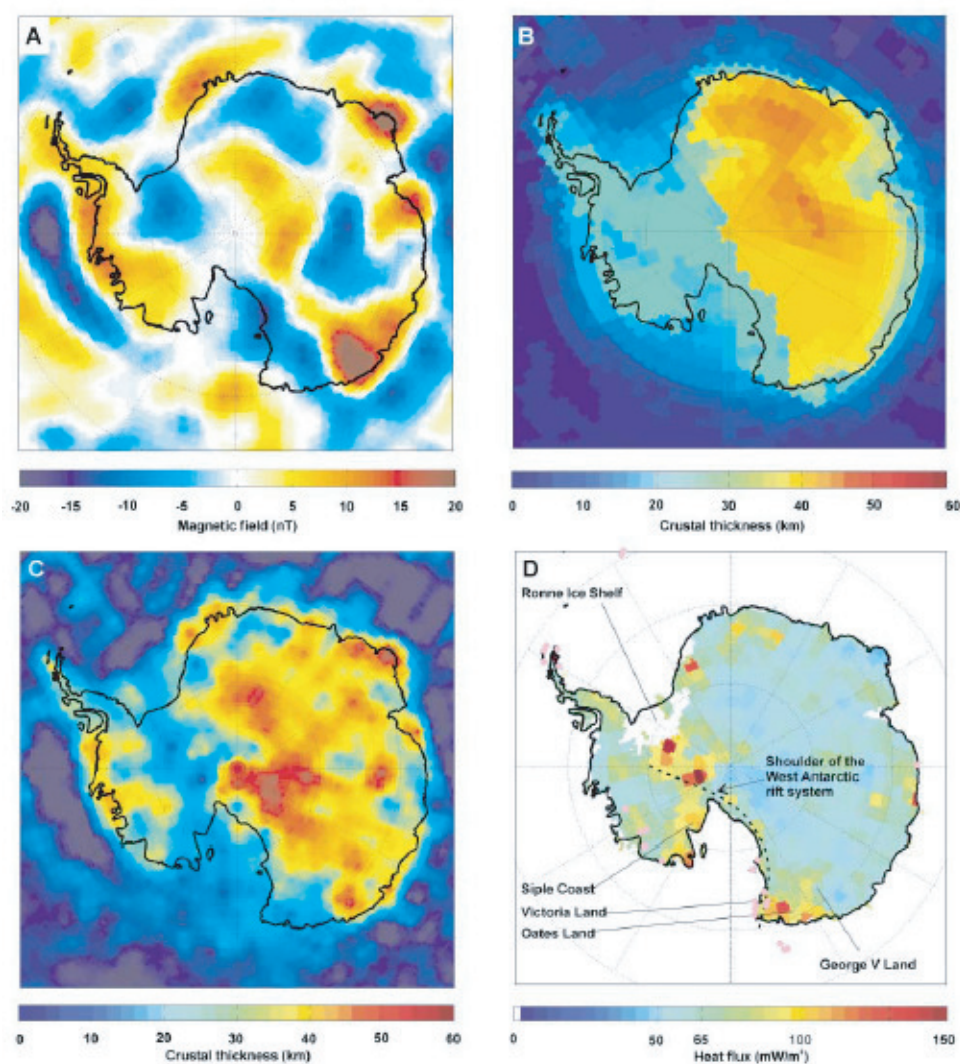
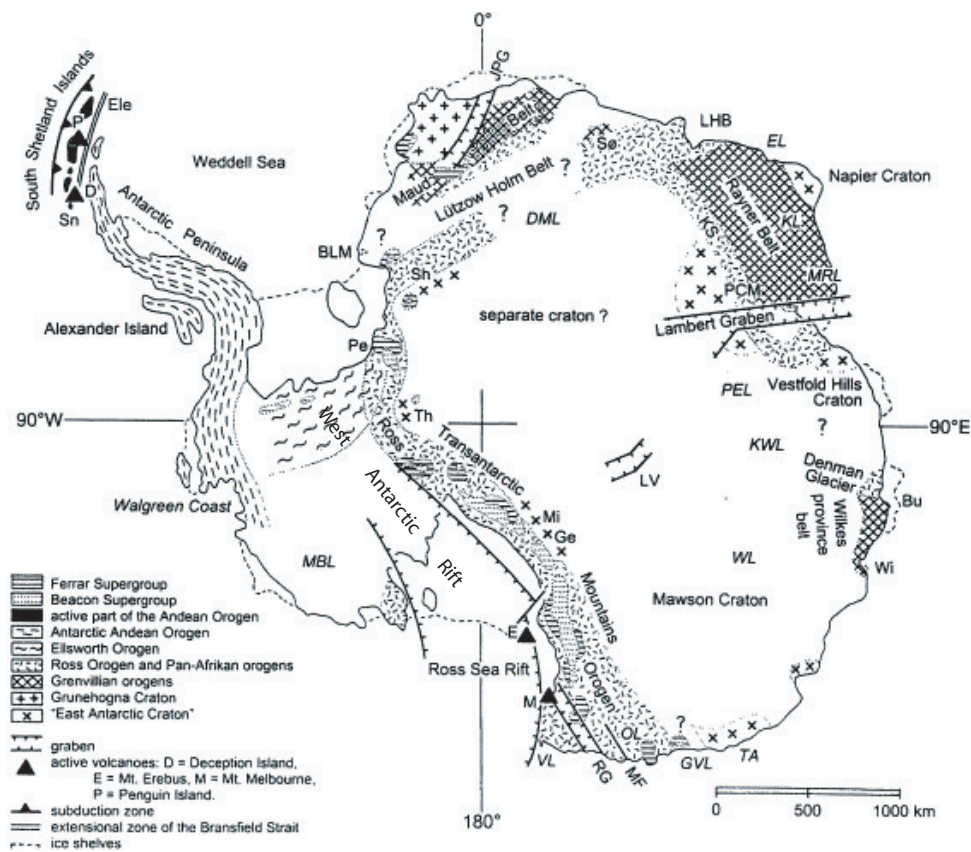


Figure 1: Estimating the Curie depth from vertical magnetic data. (a) Vertical magnetic field. (b) Crustal thickness model of Antarctica. (c) Estimate to base of magnetized layer. (d) Estimated heat flow. Figure from Maule et al. (**Science**, 2005).



Talarico and Kleinschmidt, in **Antarctic Climate Evolution**, 2008

Figure 2: Tectonic map of Antarctica from Talarico and Kleinschmidt (**Antarctic Climate Evolution**, 2008).